

Learning Equilibrium in Rank-Order Tournaments

A Reinforcement Learning Framework

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Contests are everywhere!

Different settings, same strategic structure

ASPEN GROVE
**UGLIEST
DOG CONTEST**
A FACE ONLY EVERYONE WOULD LOVE!



<https://www.thestrengthfeed.com/the-news-feed/sport-or-competition-athlete-verses-athletic>
<https://www.nbcnews.com/news/us-news/how-did-july-4-hot-dog-eating-contest-become-holiday-n386456>
<https://www.gogophotocontest.com/humanesocietyhbg>
<https://visittlitleton.org/event/ugliest-dog-contest/>

Contests in Business and Economics

Contest Theory

Key question
How do players behave?

Equilibrium Effort
A benchmark of strategic behavior



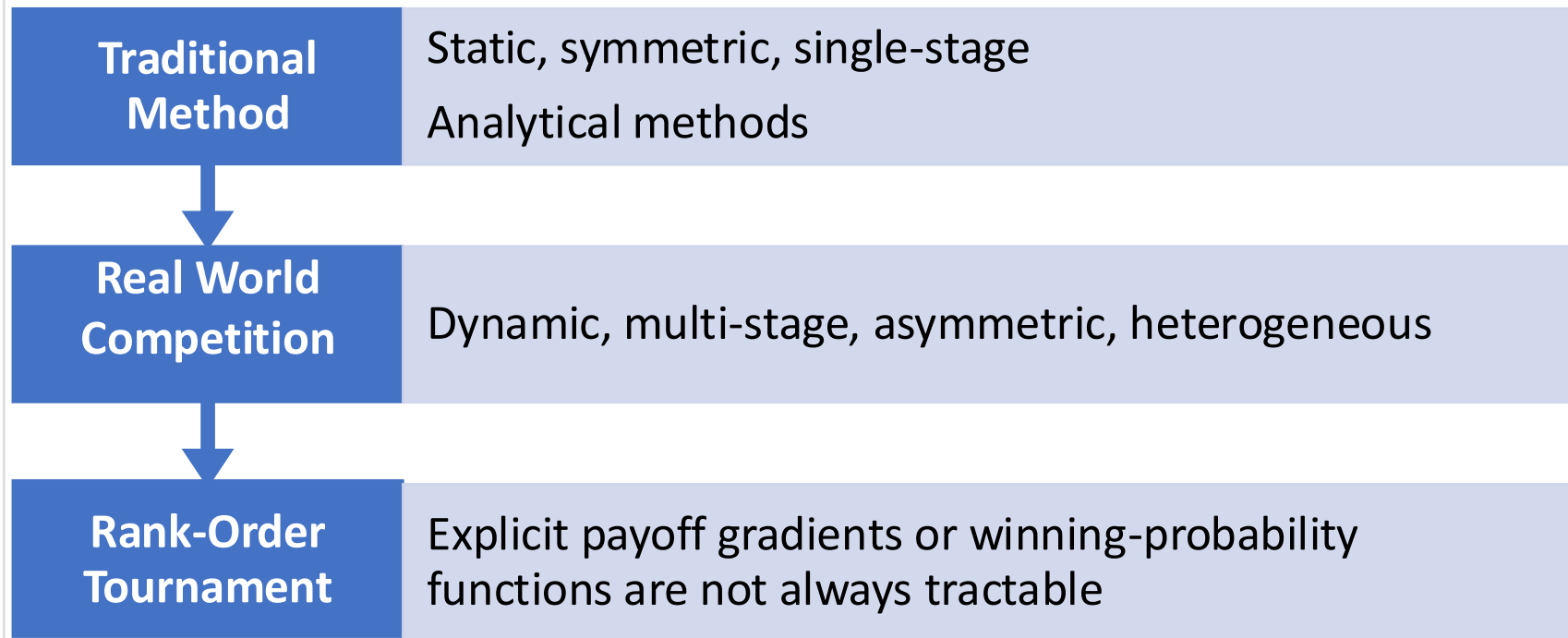
<https://corporatefinanceinstitute.com/resources/management/auction/>
<https://www.geckoboard.com/dashboard-examples/sales/sales-competition-dashboards/>

Canonical Contest Models

- Contest: Costly effort; prize competition; probabilistic success; strategic interaction
- Three canonical frameworks
 - Tullock (ratio-form) contest: $p_i = \frac{e_i^r}{\sum_j e_j^r}$
 - All-pay auction: $u_i = V \cdot \mathbf{1}\{e_i > \max e_j\} - c_i(e_i)$
 - The Lazear–Rosen rank-order tournament

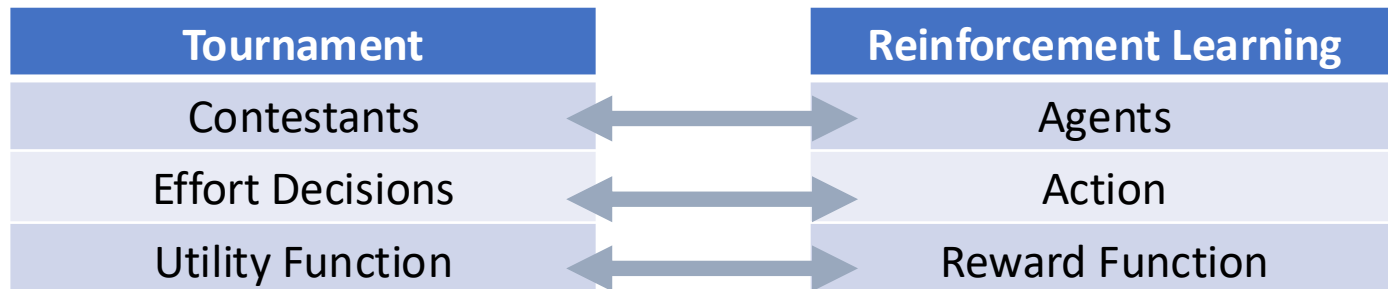
$$y_i = e_i + \varepsilon_i$$

Challenges of Existing Methods



Why Reinforcement Learning?

- Learn from outcomes
- Handle dynamic environments
- Capture Multi-agent strategic interactions
- Scale to complex settings



Motivation

- Existing learning-based work:
 - Auctions, Tullock-type contests
 - Other complex strategic environments
- Lazear–Rosen Rank-order tournaments:
Noisy performance • Discrete rang-based rewards • Intertemporal dependence
- Develop a self-play RL framework for equilibrium computation in rank-order tournaments

No closed-form winning probability or payoff gradient required

Objectives



One-stage Foundation

Develop & Validate RL equilibrium Computation



Two-stage extension

Introduce intertemporal strategic dependence



Dynamic Multi-stage Tournament

Learn & verify approximate equilibrium effort policies

**Learning Equilibrium
in One-Stage Rank-Order Tournaments**



Classical One-Stage Tournament

- Two symmetric, risk-neutral players
- Performance: $y_i = e_i + \varepsilon_i$, with $\varepsilon_i \sim U(-q, q)$
- Prize:
 - Higher Performance $\rightarrow w_H$
 - Lower performance $\rightarrow w_L$
- Cost function: $c(e_i) = ke_i^2$

One-Stage Tournament: Analytical Benchmark

- Payoff:

$$u_i = w(p_i) - c(e_i), i = 1, 2$$

- Expected Utility:

$$Eu_i(e_i, e_{-i}) = w_L + p_i(e_i, e_{-i})[w_H - w_L] - ke_i^2$$

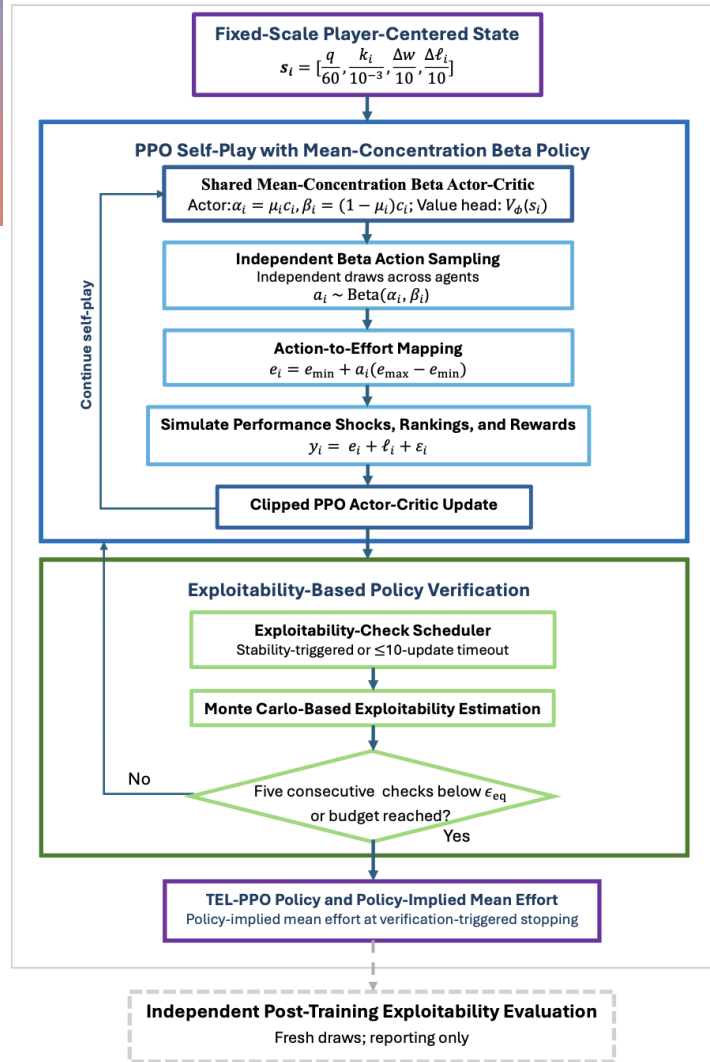
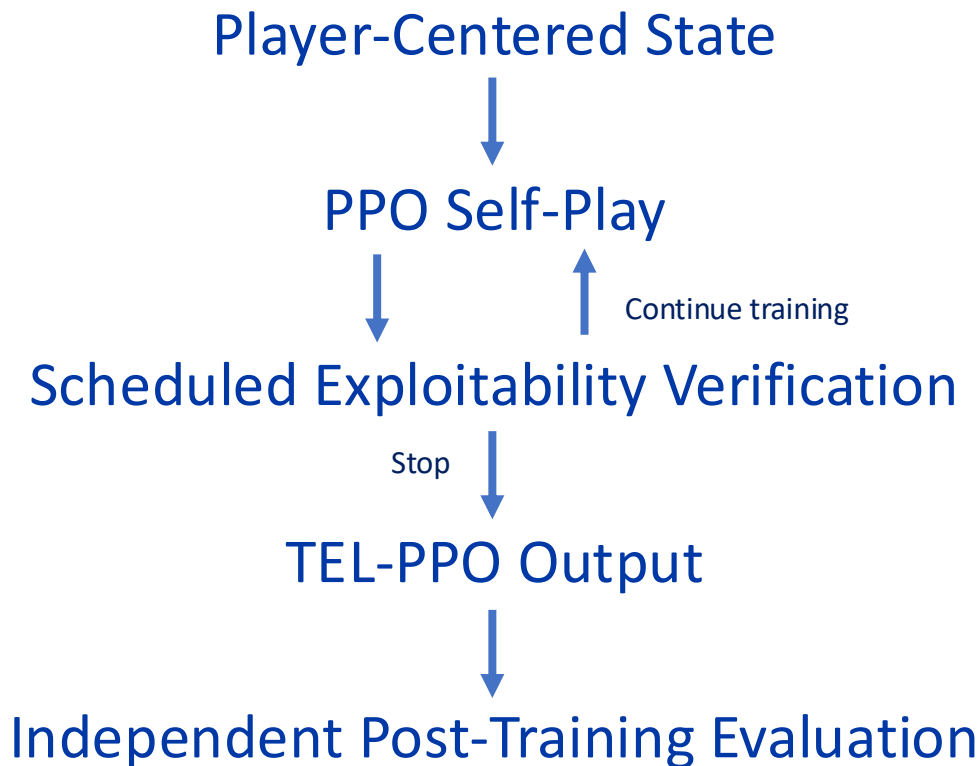
- Symmetric Two-Player Equilibrium:

$$e^* = \frac{w_H - w_L}{4qk}$$

TEL-PPO Overview

- **Learn:** PPO self-play from simulated tournament outcomes
- **Verify:** Estimated unilateral-deviation exploitability
- **Validate:** Compare with analytical equilibrium benchmark

TEL-PPO Framework



Policy Representation

- Beta Distribution ensures actions in $[0,1]$

$$a_i \sim \text{Beta}(\alpha_i, \beta_i)$$

- Mean-Concentration Beta Policy

$$\alpha_i = \mu_i c_i, \beta_i = (1 - \mu_i) c_i$$

□ μ_i : policy mean
□ c_i : concentration

- Effort mapping:

$$e_i = e_{min} + a_i(e_{max} - e_{min})$$

Tournament Outcome and Reward

- Performance: $y_i = e_i + l_i + \varepsilon_i$, with $\varepsilon_i \sim U(-q, q)$
- Realized Reward

$$r_i = \begin{cases} w_H - ke_i^2, & \text{if } y_i > y_j, \\ w_L - ke_i^2, & \text{otherwise.} \end{cases}$$

- Reported Effort

$$\hat{e}_i^{TEL} = e_{min} + (e_{max} - e_{min})\mu_{\hat{\theta}}(s_i)$$

PPO Update

- Learning Objective

$$J(\theta, \phi) = J^{clip}(\theta) - c_v L^{value}(\phi)$$

- One-step Advantage Estimate

$$\widehat{A}_i = r_i - V_{\phi}(s_i)$$

- General GAE structure retained for multistage extension
- Exploration: Stochastic Beta policy; no entropy bonus

Policy Verification

Stability Check	Exploitability Check
Has the policy stopped changing?	Can a player gain by deviating alone?
KL divergence	$Exploit_i(\pi) = \max[U_i(e'_i, \pi_{-i}) - u(\pi_i, \pi_{-i})] < \varepsilon$ π : a policy profile, U_i : expected payoff
Effort drift	Maximum unilateral -deviation gain
Schedules verification	Governs stopping

Exploitability Verification

- Training-time Verification

$$\widehat{Exploit}(\pi) \leq \varepsilon_{eq}$$

$\varepsilon_{eq} = 0.01, M_{ver} = 65,536$ draws/check,

5 consecutive successful checks $\rightarrow$ Stop

Otherwise $\rightarrow$ continue self-play

- Independent Post-training Evaluation

$$\max_i \widehat{Exploit}_i^{final}(\hat{\pi}^{TEL})$$

Fresh simulation draws, $M_{final} = 200,000$, Deviation grid: $\Delta e = 0.25$

Reporting only – no feedback to training

Experimental Scenarios

- **Two-Player** Baseline

$$e^* = \frac{w_H - w_L}{4qk}$$

- **Multi-Player** Extension: Three players, one winner

$$e^* = \frac{w_H - w_L}{4qk}$$

Experimental Scenarios - Heterogeneous Settings

- **Different Cost:** $k_1 \neq k_2$, $k_1 < k_2 \Rightarrow e_1^* > e_2^*$

$$e_1 = \frac{2k_2q(w_H - w_L)}{8k_1k_2q^2 - (k_1 - k_2)(w_H - w_L)}$$

$$e_2 = \frac{2k_1q(w_H - w_L)}{8k_1k_2q^2 - (k_1 - k_2)(w_H - w_L)}$$

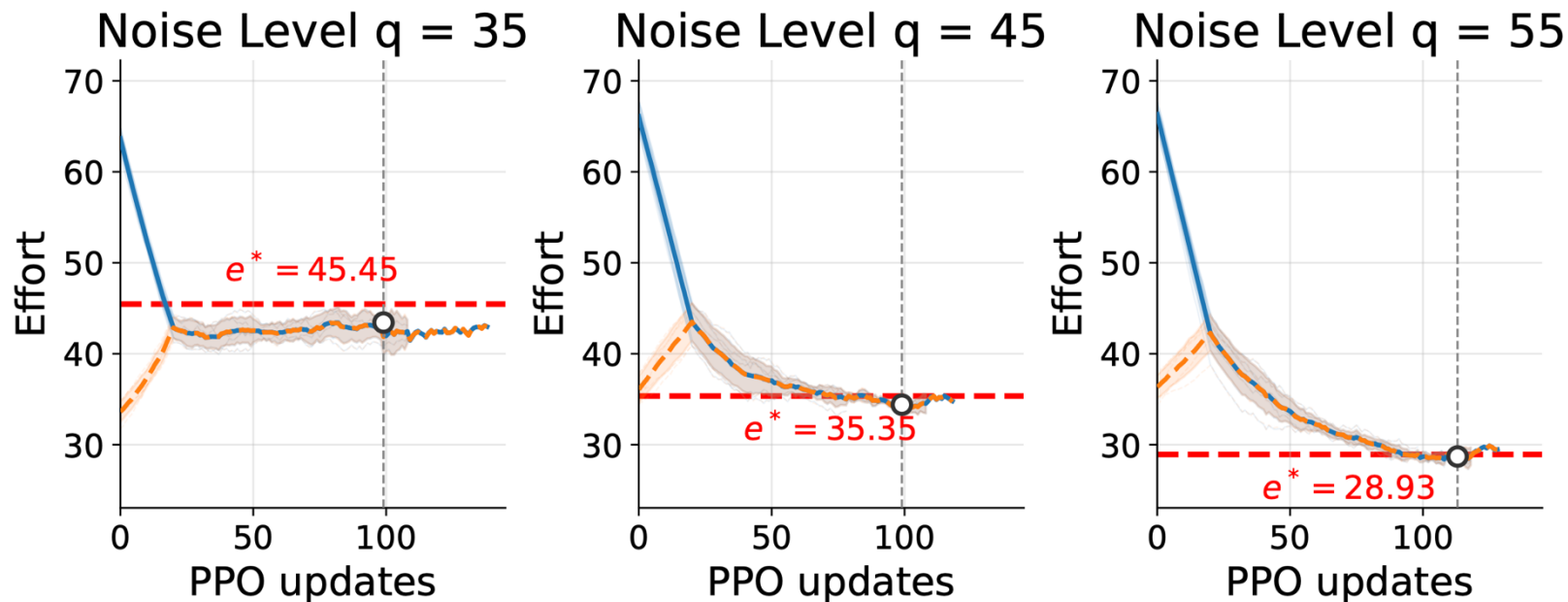
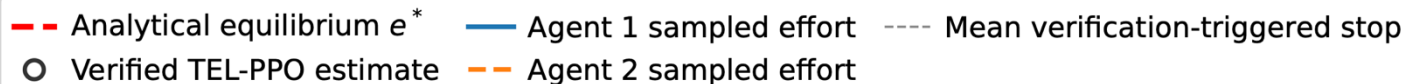
- **Different Ability:** $\ell_1 \neq \ell_2$, but a **common** equilibrium effort

$$e^* = \frac{(2q - (\ell_1 - \ell_2))(w_H - w_L)}{8kq^2}$$

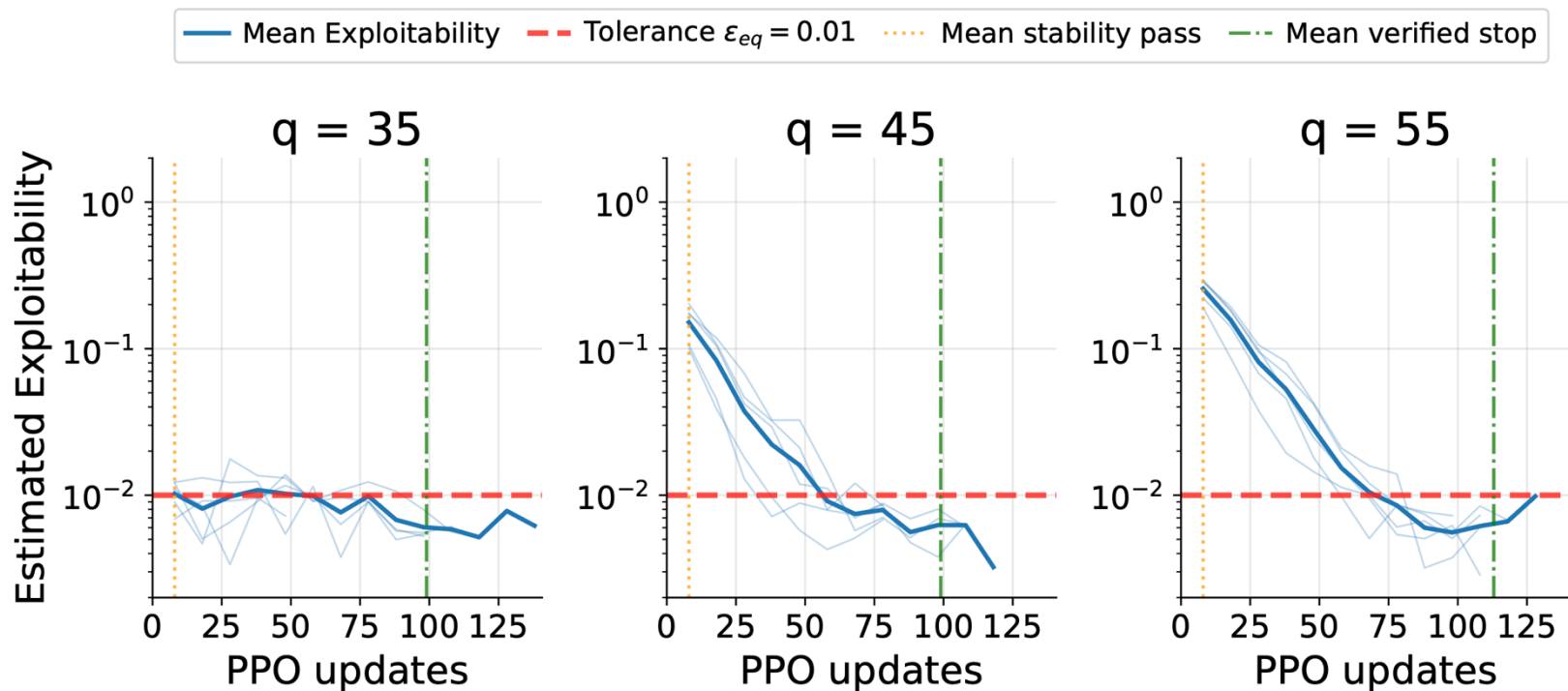
Tournament Environment Configurations

Scenario	n	Prize (w_H, w_L)	Cost / Ability	q
Two-player	2	(6.5, 3.0)	$k = 5.5 \times 10^{-4}$	35, 45, 55
Three-player	3	(6.5, 3.0)	$k = 10^{-3}$	35, 55
Heterogeneous cost	2	(8.0, 5.5)	$(k_1, k_2) = (4.0, 5.5) \times 10^{-4}$	35, 55
Heterogeneous ability	2	(6.5, 3.0)	$k = 5 \times 10^{-4};$ $(\ell_1, \ell_2) = (10, 5)$	35, 55
$e_i \in [0, 100], \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} U[-q, q]$ Common training, verification, and evaluation protocol across configurations				

TEL-PPO Effort Dynamics in the Symmetric Two-player Scenario



Training-Time Exploitability during PPO Self-play



TEL-PPO Results Across Tournament Configurations

Scenario	q	Analytical benchmark e^*	Verified TEL-PPO effort $\hat{e}_i^{\text{TEL}}$	Absolute error	Independent post-training exploitability ($\times 10^{-3}$)	MC evaluator baseline at e_i^* ($\times 10^{-3}$)
Two-player	35	45.45	43.43 ± 0.34	2.03	2.56 ± 1.16	1.22 ± 1.20
	45	35.35	34.40 ± 1.05	0.95	1.55 ± 0.99	0.74 ± 0.45
	55	28.93	28.70 ± 0.89	0.22	1.55 ± 0.79	0.61 ± 0.38
Three-player	35	25.00	24.61 ± 0.64	0.39	2.56 ± 1.43	1.51 ± 0.46
	55	15.91	17.03 ± 0.15	1.12	3.12 ± 0.72	0.64 ± 0.39
Heterogeneous cost	35	38.03 / 27.66	37.77 ± 0.95 / 27.05 ± 1.25	0.25 / 0.61	1.51 ± 1.13	0.51 ± 0.25
	55	26.54 / 19.30	25.64 ± 1.02 / 20.02 ± 1.88	0.90 / 0.72	1.82 ± 1.33	0.28 ± 0.18
Heterogeneous ability	35	46.43	45.86 ± 0.80	0.57	2.04 ± 1.63	0.88 ± 0.54
	55	30.37	31.09 ± 0.81	0.71	1.28 ± 0.92	0.85 ± 0.41

- **Effort recovery:** MAE=0.77; max absolute error=2.03
- **Cost heterogeneity:** correct effort ordering recovered
- **Ability heterogeneity:** common benchmark recovered

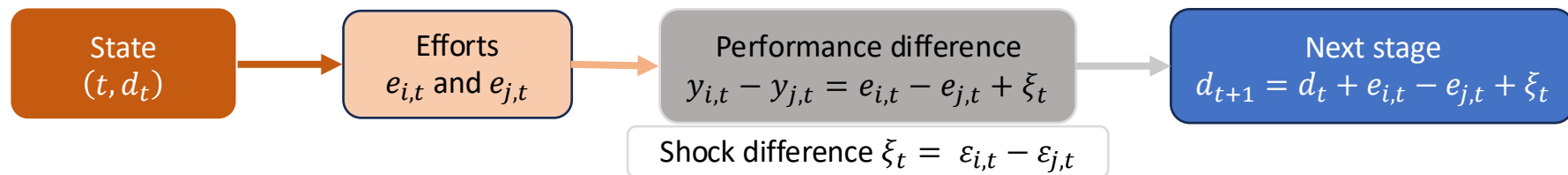
Conclusion

- Recovery of known equilibrium effort patterns
- Distinct effort responses under cost heterogeneity
- Low estimated unilateral-deviation gains after training
- A clean separation among learning, stopping, and final evaluation

Learning and Verifying Approximate Equilibria in Dynamic Rank-order Tournaments



TEL-PPO From One-stage to Multistage



- Game structure: Static $\rightarrow$ Dynamic with intermediate states
- Equilibrium object: $e^* \rightarrow$ State-dependent policy $e_t(d_t)$
- Cumulative score gap

$$d_t = S_{i,t-1} - S_{j,t-1} = \sum_{\tau=1}^{t-1} y_{i,\tau} - \sum_{\tau=1}^{t-1} y_{j,\tau}$$

Markov Strategy and Equilibrium

- Sufficient Markov state: Current stage + cumulative score gap

$$s_t = (t, d_t)$$

- Continuation value: Future value from the current state

$$V_t(d_t)$$

- Approximate Markov Perfect Equilibrium: Optimal at every payoff-relevant stage and state

$$\text{Approximate MPE: } G_{\max}^{\text{full}} \leq \varepsilon$$

Backward Stagewise Learning



- Active stage: stochastic PPO
- Later stages: frozen continuation policies
- Earlier stages: include expected continuation value

$$E_{\xi_t}[V_{t+1}^{frozen}(d_t + e_{i,t} - e_{j,t} + \xi_t)]$$

Retained and New

Component	One-stage TEL-PPO	Multistage Extension
Policy representation	Mean-concentration Beta	Retained
Self-play	TEL-PPO framework	Retained
Equilibrium object	Scalar effort	State-dependent policy
Training	Direct self-play	Backward stagewise+ frozen continuation
Verification	Static unilateral deviation	Dynamic best response

Two-Stage Model as a Validation Environment

- **Policy recovery**

- $e_1^*(0) = \frac{w^H - w^L}{6qk}$

- $e_2^*(d) = \frac{\Delta W}{8\kappa q^2} (2q - |d|)_+$

- Benchmark: $e_1^*(0) = E[e_2^*(d_2)]$

- **Algorithm Validation**

- **Verifier calibration**

Dynamic Best-Response Verification

- Root Exploitability: Dynamic best-response gain from the initial state
- Reachable-state diagnostic: One-step deviation incentives over BR-reachable states
- Full-domain verification: Dynamic best-response gain over all feasible states

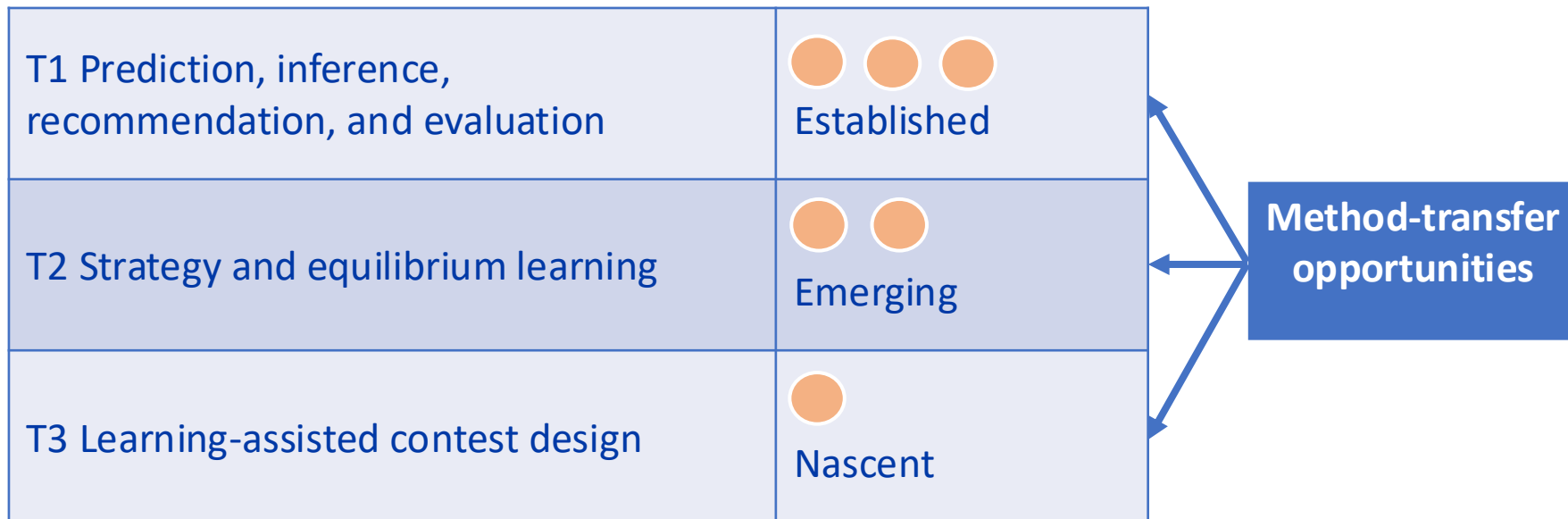


Full-domain MPE-oriented verification (grid-based)

How to Evaluate the Verifier

- Test 1: Analytical-equilibrium calibration:
 - Known equilibrium → **near-zero** deviation gain
- Test 2: Controlled policy misspecification
 - Perturbed policies → **larger** detected deviation gains
- Test 3: Numerical refinement:
 - Finer grids / integration → **stable** verification results

Learning-based methods in Contests: Current Landscape



Future Research



Richer dynamic
equilibrium learning



Competitive behavior
in the dynamic
tournaments



Learning-assisted
Contest Design



Questions & Comments?

Thank you!